

A number of important insights can be gleaned from this chart. First, notice that the yield, which once again is the annualized income stream divided by the initial premium, is much greater than the yield on a government bond. The reason for this is because your income is an integrated blend of three distinct cash flows: First, you are getting a type of interest coupon on your money; second, you are getting a portion of your premium back, and third—the main benefit derived from the annuity—you are getting some of other people's money.

What do I mean by “other people's money”? This is what I earlier referred to as mortality credits, and here is a simple tale that can illustrate this concept, as well as the benefits of annuitization and longevity insurance.

Yet Another Digression—The Tontine

Imagine a group of five healthy 95-year-old females, all coincidentally celebrating their 95th birthday on the same day. To bring some excitement to the party, they decide to engage in an odd gamble. Each of the five women agrees to contribute \$100 to a pool. The funds will be frozen for a year, they decide. “Whoever survives to the end of the year gets to split the \$500; whoever doesn't make it, forfeits the money.” This type of arrangement is known as a tontine, and is named after Lorenzo Tonti, an Italian banker who first promoted the concept to King Louis XIV of France around the year 1650.

While they wait for the next year to determine the winners and losers from the tontine, the five of them decide to put the money in a local bank's one-year certificate of deposit (CD) paying 5% interest for the year. The CD will mature at \$525 at year end, just in time for their 96th birthday.

So what exactly will happen next year? Who will survive? How much will they get? Well, according to statistics compiled by actuaries at the U.S. Social Security administration, there is roughly a 20% chance that any given 95-year-old will die during the next year. This, in turn, implies an 80% chance of survival for any one of them. These odds imply that we expect four out of five women to survive and make it to their 96th birthday to split the \$525 pot at year-end.